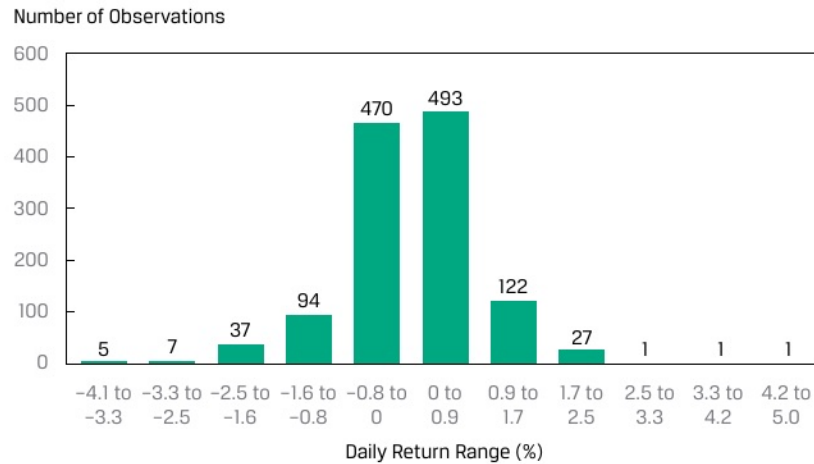


Exhibit 1: Histogram of Daily Returns on the EAA Equity Index

Measures of Central Tendency

The Arithmetic Mean

Analysts and portfolio managers often want one number that describes a representative possible outcome of an investment decision. The arithmetic mean is one of the most frequently used measures of central tendency.

Arithmetic Mean. The **arithmetic mean** is the sum of the values of the observations in a dataset divided by the number of observations.

The Sample Mean

The sample mean is the arithmetic mean, or arithmetic average, computed for a sample.

Sample Mean Formula. The **sample mean** or average, $\bar{X}$ (read “X-bar”), is the arithmetic mean value of a sample:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}, \quad (1)$$

where n is the number of observations in the sample.

A property and potential drawback of the arithmetic mean is its sensitivity to extreme values, or outliers. Because all observations are used to compute the mean and are given equal weight (i.e., importance), the arithmetic mean can be pulled sharply upward or downward by extremely large or small observations, respectively. The most common approach in this situation is to report the median, or middle value, in place of or in addition to the mean.

The Median

A second important measure of central tendency is the median.

Definition of Median. The median is the value of the middle item of a dataset that has been sorted into ascending or descending order. In an odd-numbered sample of n observations, the median is the value of the